

Chapter II

Cutaneous Mycoses

Cutaneous or dermatophytoses mycoses

- Caused by fungi that infect **only** the keratinized tissue (skin, hair, and nails).
- A group of 40 related fungi that belong to three genera:
 - *Microsporum*, hair and skin
 - *Trichophyton*, hair, skin, and nail
 - *Epidermophyton*, skin and nails
- These fungi all have the ability to degrade keratin and grow as noninvasive saprotrophs on skin and its appendages
- The dermatophytes grow on the non-living outer layers of the skin (**stratum corneum**), and only exceptionally invade living tissue.
 - Because most are unable to grow at **37°C** or in the presence of serum.

- The dermatophytes are the **only fungi** that have evolved a **dependency** on human or animal infection for the survival and dissemination of their species.
 - Produce two kinds of conidia:- macroconidia & microconidia
 - Resistant to cycloheximide
- Unlike superficial mycoses, dermatophyte infections can be itchy, inflammatory, and affect the smooth skin, hair, and nails.
- Dermatophytes produce extracellular enzymes (keratinases) capable of hydrolyzing keratin.

- The dermatophytoses are often distinguished according to their natural habitats:
 - Anthropophilic (*E. Floccosum*, *T. rubrum* etc.)
 - Zoophilic (*T. verrucosum*, *M.canis* etc.)
 - geophilic (*M. cookei*, *M. gypseum* etc.)
- Transmission: is by infected skin scales on inanimate objects. Only the pathogenic fungi are capable of human to human spread.
- Cause common tinea (ringworm), which is a very common fungal infection of the skin.
- Tinea is often called "ringworm" because the rash is circular, with a ring-like appearance

Clinical presentation

- Clinically, ringworm can be classified depending on the site affected
 - *Tinea capitis* involving scalp
 - *Tinea corporis* involving non-hairy skin of the body
 - *Tinea cruris* affecting groin
 - *Tinea pedis* affecting foot
 - *Tinea barbae* affecting beard areas of face and neck

Tinea capitis

T. tonsurans
T. soudanense
T. schoenleinii
T. violaceum
T. verrucosum
M. audouinii
M. canis

Tinea imbricata

T. concentricum

Tinea cruris

T. rubrum
E. floccosum

Tinea manuum

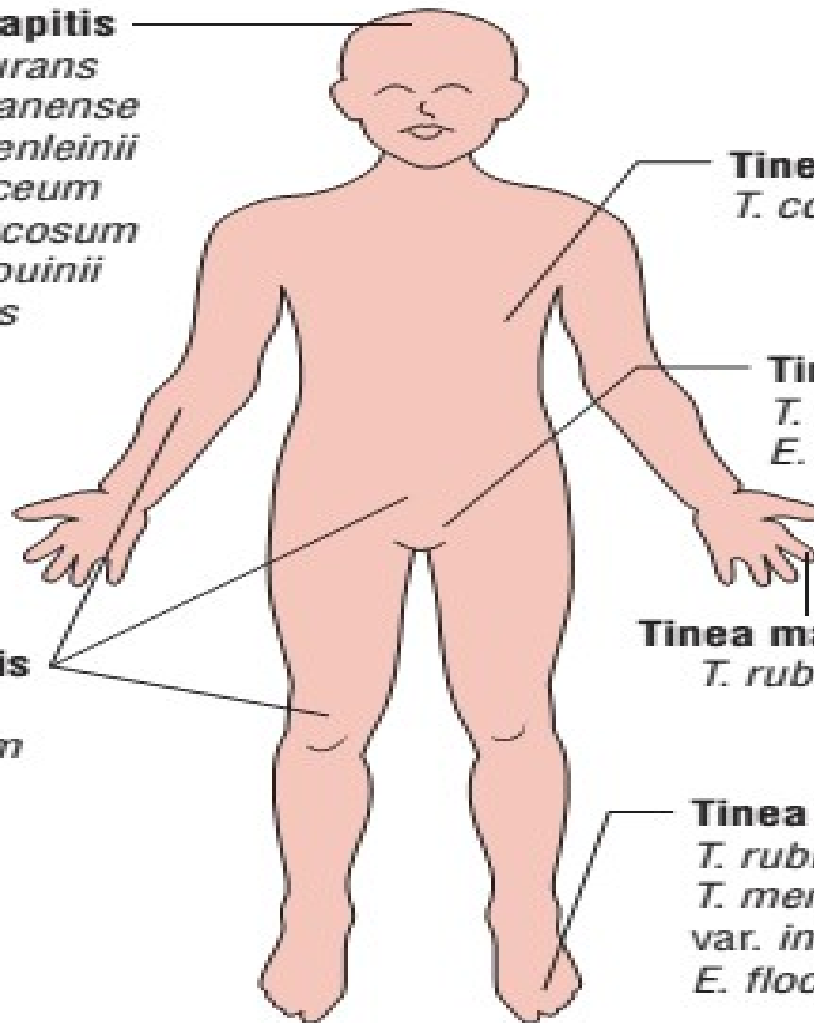
T. rubrum

Tinea corporis

T. rubrum
T. verrucosum
M. gypseum
M. canis

Tinea pedis

T. rubrum
T. mentagrophytes
var. *interdigitale*
E. floccosum



Laboratory diagnosis

- **Specimen**, Skin scraping, nail clipping, plucked hair
- **Microscopy**
 - 10-25% KOH - hyphal elements; **arthroconidia**
 - branching hyphae or chains of arthroconidia (arthrospores) are seen
 - Spores of *Microsporum*-infected hairs fluoresce under Wood's light in a darkened room.
- **Culture**
 - Inhibitory mold agar or SDA containing cycloheximide and chloramphenicol, incubate at 25 C.
 - Species are identified on the basis
 - colonial morphology (growth rate, surface texture, and any pigmentation),
 - microscopic morphology (macroconidia, microconidia), and,
 - in some cases, nutritional requirements

SUBCUTANEOUS MYCOSES

- Fungal infections of the **dermis**, **subcutaneous tissue**, and **bone**.
 - deeper layers of the skin, including the cornea, muscle, and connective tissue
- Caused by a broad spectrum of taxonomically diverse fungi and introduced by trauma
- Localized lesions, slowly progressive
- There are three general types of subcutaneous mycoses
 - Chromoblastomycosis,
 - Eumycotic mycetoma
 - Sporotrichosis

CHROMOBLASTOMYCOSIS

- Granulomatous fungal infection
- pigmented fungi(dematiaceous)
- Sclerotic or muriform bodies or copper penny bodies
- Adult male agricultural workers
 - Fonsecaea Species
 - Phialophora verrucosa
 - Cladosporium carrionii
 - Fonsecaea compacta
 - Rhinocladiella aquaspersa- rare

EUMYCOTIC MYCETOMA

- bone and lymph nodes
- multiple granulomas and abscesses
 - fungal hyphae known as **granules** or **grains**
 - Tumefaction
 - draining sinuses
- Causes gross deformity
- more common in men
- Pseudallescheria boydii
- Nocardia brasiliensis

Sporotrichosis

- Occupational disease of gardeners and farmers
- a granulomatous ulcer at the puncture site
- deficiencies in T-cell function
 - Dissemination to distant sites
- **Sporothrix schenckii**(dimorphic)
- **asteroid body** amorphous eosinophilic “rays.”
- Cutaneous and extra-cutaneous
 - Pulmonary, localized and disseminated

Key concepts of Subcutaneous Mycoses

- Caused by dozens of environmental molds associated with vegetation and soil.
- Acquired usually when minor cuts or scratches introduce soil or plant debris containing the pathogenic fungus.
- *Sporothrix schenckii*, the cause of sporotrichosis, is a dimorphic fungus that converts from hyphal growth to yeast cells
- The diagnostic feature of chromoblastomycosis is the microscopic observation of brownish (melanized), spherical sclerotic bodies within the lesions.
- The hallmark of a mycetoma is localized swelling and the formation of fistulae that contain hard granules composed of hyphae and inflammatory tissue

Laboratory Diagnosis

- Samples: aspirated material, biopsy tissue
- Direct microscopic examination (KOH),
 - elongated conidiospores (*S. schenckii*)
 - thick walled dark brown sclerotic bodies or muriform cells
 - broad septate hyphae with well defined walls and chlamydospores *Pseudallescheria boydii*
 - Histopathological examination (methenamine silver stain)
 - **asteroid body**
- **Culture**
 - SDA with and without antibiotics
 - Typical velvety brown colored colony

Systemic or Endemic Mycoses

- Disseminated infections that affect internal organs or deep tissues
- The organisms responsible for systemic mycoses fall into two general categories:
 - true pathogens
 - opportunistic pathogens
- Inhalation  pulmonary infection  dissemination
- The causative agents are Dimorphic fungi
 - a mold form in 25°C to 30° C and a yeast or spherule form in tissues or when grown on enriched medium in the laboratory at 37° C
- 90% resolve in immunocompetent individuals

⌄ The dimorphic pathogens include

- *Blastomyces dermatitidis*, **Blastomycosis**
- *Coccidioides immitis* and *C. posadasii*, **Coccidioidomycosis**
- *Histoplasma capsulatum*, **histoplasmosis**
- *Paracoccidioides brasiliensis*, **Paracoccidioidomycosis**
- *Penicillium marneffe*

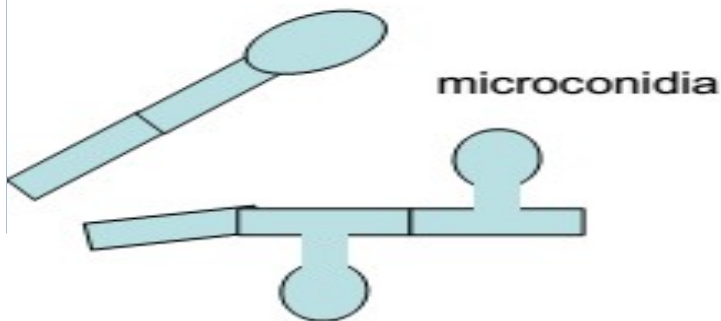
⌄ Systemic mycoses occur in three disease forms:

- acute self limited pulmonary (asymptomatic to severe),
- chronic (pulmonary or disseminated)
- disseminated.

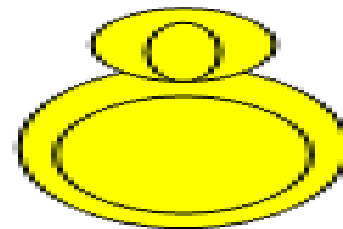
Blastomyces dermatitidis

- ❑ Thermally dimorphic fungus that grows
 - As a mold producing hyaline, branching septate hyphae and conidia.
 - Non-encapsulated yeast like cells in tissue and in culture on enriched media at 37°C, Buds are produced singly and are attached to parent cell by broad base.

Mold phase



Tissue phase

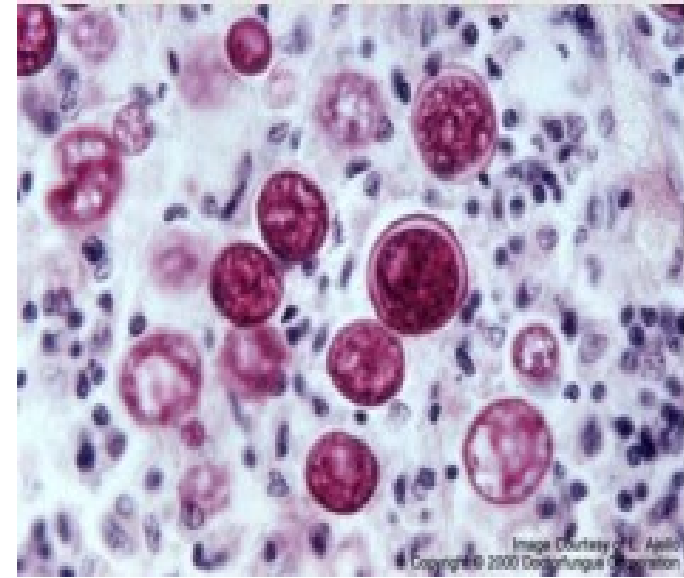


□ 2 phases: asexual & sexual

➤ Blastomyces dermatitidis
(asexual phase)

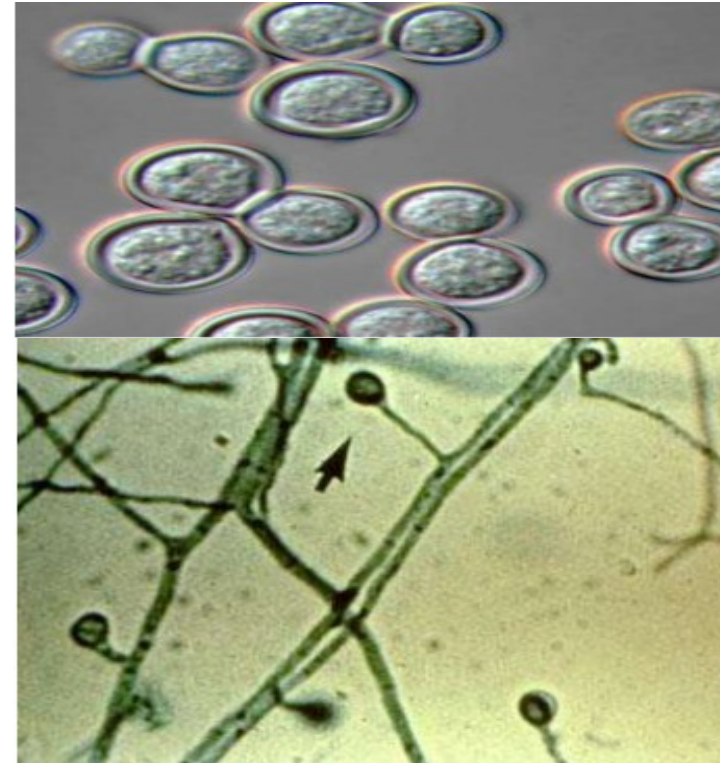
➤ Ajellomyces dermatitidis
(sexual phase)

□ Dimorphic fungus



Laboratory diagnosis

- **Specimen:** sputum, pus, biopsy
- **Direct microscopy:** 10% KOH, H&E
 - **Yeast form:** Blastoconidia attached by broad base
 - **Mold phase:** Lollipop forms
- **Culture**
 - *represent a biohazard*
 - Saborauds, Mycosel or Mycobiotic Agar
 - Two media inoculated
 - producing a fluffy white mycelium or slowly as glabrous, tan, non sporulating colonies



Coccidioidomycosis

□ 2 species:

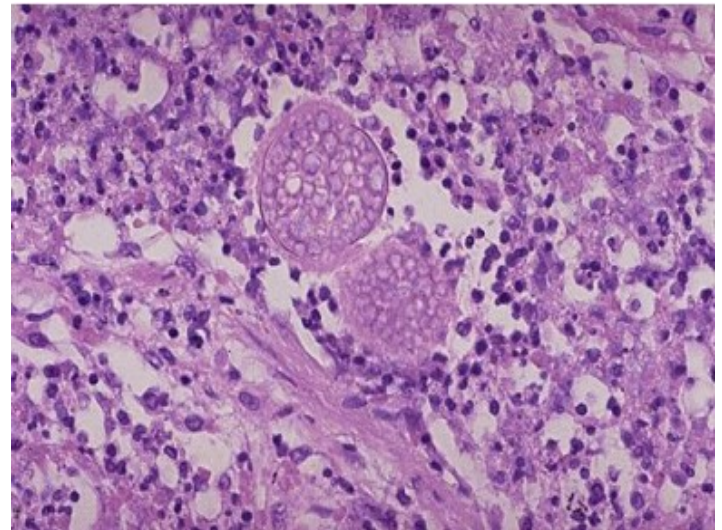
- *C. immitis* and *C. posadasii*
- Phenotypically indistinguishable
 - Genotyping and phylogenetic analyses
- Dimorphic that grows as a **mycelium** in the soil and produces a **spherule** form in the host organism
- The disease is usually mild, with flu-like symptoms, cough, myalgia, fever, headaches and rashes.
- Caused by the inhalation of infectious **arthroconidia**

Clinical features

- Majority – asymptomatic
- Pulmonary disease which resembles mild influenza like fever
(Valley fever) to Pneumonia
- Dissemination- uncommon but highly fatal

Laboratory diagnosis

- **Specimens;** Sputum, Pus, Biopsy
- **Direct examination:** KOH; H&E
 - **Yeast form:** thick double walled refractile a spherule filled with endospores
 - **Mycelial form:** Pseudohyphae which fragments into arthrospores highly infectious
- **Culture**
 - SDA or BHI agar with cycloheximide and chloramphenicol
 - When working with known *Coccidioides*, Biosafety Level 3 is required



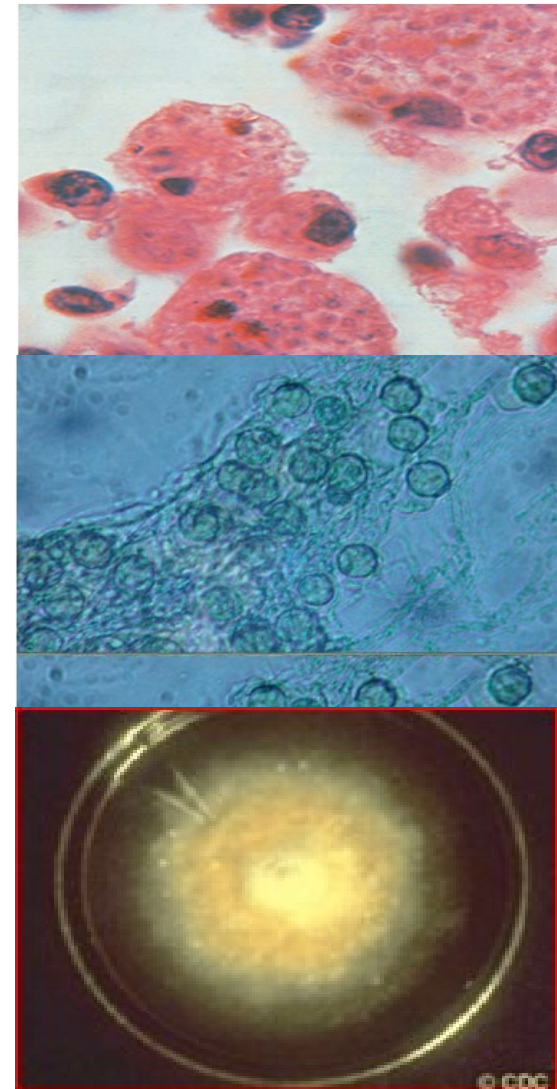
Histoplasma capsulatum

- Two varieties
 - var. *capsulatum* & var. *duboisii*
- **var. *capsulatum***
 - **Disease:** HISTOPLASMOSIS
 - **Natural reservoir:** soil, bat and avian habitats
- **var. *duboisii***
 - African Histoplasmosis
 - Differentiated from classical histoplasmosis by
 - Larger, thick-walled yeast cells
 - Pronounced giant cell formation in infected tissue
 - Diminished pulmonary involvement
 - Greater frequency of skin and bone lesions
 - indistinguishable in their mould forms

- Dimorphic soil saprophyte
- **Mycelial phase**: 25° C; Macroconidia (8-14 µm), Microconidia (2-4 µm)
- At 37°C or at body temperature = **budding yeast** 2-3 x 3-4 µm
Found predominantly in histiocytes,
 - yeast cells of both varieties are **intracellular** in vivo and are uninucleated
- most prevalent pulmonary fungal infection in humans and animals
- initiated by inhalation of the **conidia** from Soil enriched with excreta of birds or bats

Laboratory diagnosis

- **Samples:** Sputum, tissue, bone marrow, CSF, blood
- **Direct examination:** Giemsa / Wright
 - small oval yeast cells, packed within the cytoplasm of macrophages or monocytes
- **Culture**
 - **SDA:** white or buff-brown, suede-like to cottony with a pale yellow-brown
 - *severe biohazard*



Paracoccidioides brasiliensis

- Dimorphic
 - grows as a mycelium that produces chlamydoconidia, terminal conidia and arthroconidia infectious form
 - yeast forms displaying several blastoconidia (forming a “pilot’s wheel”)
 - Both morphotypes require ample oxygen supply for their growth
- Paracoccidioidomycosis
- the mold form grow very slowly
- causes a granulomatous or ulcerative lesion in the GIT

Penicillium marneffe

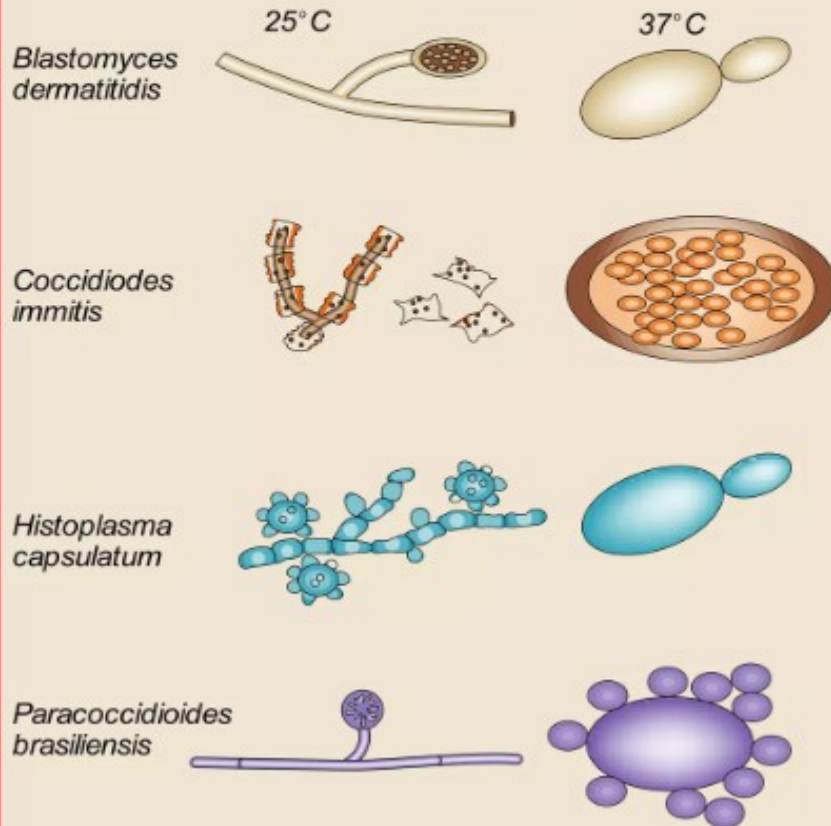
- *Talaromyces marneffe*
- causes life-threatening disseminated infection
- intracellular pathogen that exudes a brilliant **red pigment** into the agar around the colony
- infection is acquired by inhalation of conidia
- dimorphic pathogen
 - filamentous in environmental conditions and yeast at body temperature

Key Concepts of Endemic Mycoses

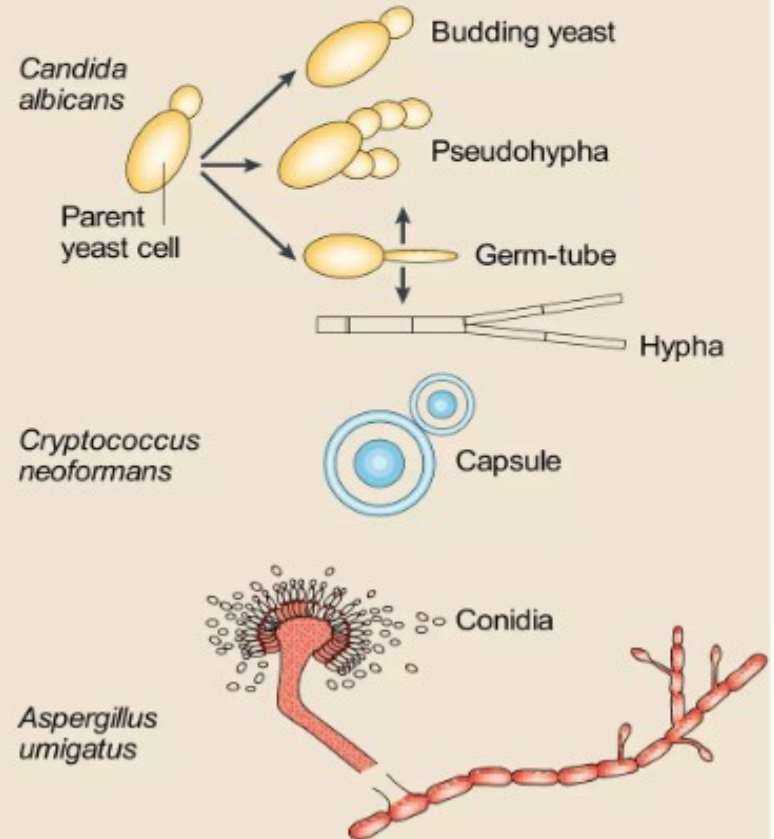
- ➡ Characterized by distinct geographic distribution and are dimorphic
- ➡ Initiated by inhaling **conidia** of the causative fungi and within the lungs convert to distinctive yeast cells
- ➡ Individuals with compromised CMI defenses have a significantly greater risk of developing disseminated disease
- ➡ The incidence of disseminated disease is significantly higher in males.
- ➡ Serologic tests for serum antibodies to the endemic fungi have diagnostic and prognostic value

Medically Important Fungi

Dimorphic fungi



Opportunistic fungi



Opportunistic Systemic Mycoses

- Fungal infections of the body occur
 - **debilitated** patients whose normal defense mechanisms are impaired
- The organisms have a very **low inherent** virulence
- The increased incidence of these infections and the diversity of fungi causing them
 - emergence of AIDS
 - more aggressive cancer and post-transplantation chemotherapy
 - use of antibiotics, cytotoxins, immunosuppressives, corticosteroids
- **Endogenous** and **exogenous** fungi

Opportunistic Systemic Mycoses

Disease	Causative organisms	Incidence
Aspergillosis	Aspergillus fumigatus complex	Common
Candidiasis	Candida	Common
Cryptococcosis	Cryptococcus spp (especially C. neoformans and C. gattii.)	Common
Zygomycosis (Mucormycosis)	Rhizopus, Mucor, Rhizomucor, Lichtheimia etc	Rare

Key concepts of Opportunistic Mycoses

- Caused by globally distributed fungi
 - either members of the human microbiota or environmental yeasts and molds
- Innate host defenses (eg, neutrophils, monocytes) provide crucial protection
 - hematologic dyscrasias and immunosuppressive therapy
- Successful management depends involves
 - early diagnosis
 - rapid administration of appropriate antifungal therapy
 - control of the underlying condition or disease

Antifungal Agents

- Antifungal drugs are those drugs which inhibit or retard fungal growth.
 - fungicidal or fungistatic.
 - antibiotic or synthetic compounds.
- Antifungal drugs target structures or functions that are necessary in fungal cells but not in human cells
 - cell membrane and the fungal cell wall
- Antifungal agents currently available for clinical use belong to three major classes:
 - polyenes, azoles and 5-fluorocytosine

- **Polyenes**

- natural products of *Streptomyces nodosus*
- Amphotericin B and Nystatin
- bind to cell membrane ergosterol and disrupt their structure and function
- Active against *Cryptococcus* spp, most *Candida* spp and *Aspergillus* spp
- active against the dimorphic fungi, including
 - *H. capsulatum*, *B. dermatitidis*, *C. immitis* and *Paracoccidioides* spp
- renal toxicity, infusion reactions, electrolyte abnormalities, and hepatotoxicity

- **Azoles**
 - synthetic drugs; imidazoles, and triazoles.
 - act on ergosterol biosynthesis by inhibiting cytochrome P450 14 α demethylase
 - lanosterol to ergosterol (the main sterol in fungal cell membrane)
 - resulting in ergosterol depletion and accumulation of methylated sterols
 - Used topically or systematic (oral route only) to treat
 - Oral & vaginal candidiasis, Dermatophytosis and Systemic mycoses
 - Hepatotoxic, Inhibits human P 450 enzymes, Inhibits adrenal & gonadal steroids

- **Flucytosine**

- a fluorinated pyrimidine (5-fluorocytosine)
- impairs nucleic acid synthesis, ultimately interfering with protein synthesis as well
- often given in combination with amphotericin B & itraconazole
- *Candida* spp., *C. neoformans*, *Rhodotorula* spp., *Saccharomyces cerevisiae*
- Flucytosine is relatively nontoxic
 - Bone marrow suppression, hepatotoxicity, and gastrointestinal intolerance

❑ Fungi, like mammalian cells, are eukaryotic cells

- Antifungal agents can indiscriminately disrupt the normal metabolic processes of both fungal and host cells
- development of toxic side-effects
- Insolubility or decreased solubility of many antifungal agents
- poor absorption through the gastrointestinal tract demands the use of parenteral routes
- grow slowly and often in multicellular forms, they are more difficult to quantify than bacteria